
“The Ultrametric Sieve: A Consilience of Number, Nature, Mind, and Form”

Rowan Quni-Gudzinis | 2026-06-19

Abstract

We propose a foundational shift in the representation of integers: from decimal digit-strings to valuation vectors recording prime exponents. This “ultrametric sieve” reveals that the integers form an infinite-dimensional lattice whose geometry is the product of p -adic Bruhat–Tits trees. We trace the consequences of this shift across number theory (adèles, Riemann zeta, explicit formula, abc conjecture), physics (spin glasses, holographic duality, quantum chaos), technology (cryptography, error correction, signal processing), cognitive science (innate divisibility sense, sacred numbers), monetary economics (optimal denominations, financial time), music (just intonation, timbre, rhythm), and philosophy (Laws of Form). The valuation vector emerges as a universal Rosetta Stone—a single coordinate system in which the deepest structures of mathematics, mind, and physical reality are seen to be one structure.

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Part I — The Foundational Shift

1. The Copernican Move

We begin with a deceptively simple move: instead of reading a number by its decimal digits, read it by its prime factorisation. For the sacred Tibetan Buddhist number

108, the decimal string “1-0-8” is an arbitrary cultural artefact. Its invariant fingerprint is the valuation vector

$$v(108) = (v_2, v_3, v_5, v_7, \dots) = (2, 3, 0, 0, \dots)$$

recording that $108 = 2^2 \cdot 3^3$. This vector is completely independent of any numeral base. It is the true, irreducible fingerprint of the integer.

The ultrametric sieve is the systematic extraction of this vector for every integer. Contrast it with the classical Sieve of Eratosthenes, which answers a single Boolean question: “prime or composite?” The valuation sieve answers: “What is the complete multiplicative anatomy?”

2. Why “Ultrametric”

The p -adic absolute value

$$|n|_p = p^{-v_p(n)}$$

satisfies the strong triangle inequality $|x + z|_p \leq \max(|x - y|_p, |y - z|_p)$,

making it an ultrametric. The sieve extracts the ultrametric data for all primes simultaneously.

Part II — The Mathematical Edifice

3. The Integers as an Infinite-Dimensional Lattice

Each prime is an independent axis. The integers are points in $\mathbf{N}^\infty$

with coordinates

$$v(n) = (v_2, v_3, v_5, \dots)$$

. Multiplication is coordinate-wise

addition:

$$v(a \cdot b) = v(a) + v(b)$$

GCD and LCM become coordinate-wise min and max — the divisibility lattice is a distributive ultrametric lattice.

4.

p

p -adic Geometry and the Bruhat–Tits Trees

For each prime

p

, the

p

p -adic metric defines a regular tree (the Bruhat–Tits

tree) whose boundary is $\mathbf{Q}_p$

$\mathbf{Q}_p$

. The valuation $v_p(n)$

is the depth of n

in this tree — the distance from the root. The full space of integers is the product of all such trees.

5. The Adèle Ring and the Chinese Remainder Theorem

The Chinese Remainder Theorem $\mathbf{Z}$ embeds

diagonally into the profinite

completion

$$\hat{\mathbf{Z}} = \prod_p \mathbf{Z}_p$$

. Adding the real place yields

the adèle ring

$$\mathbf{A}_{\mathbf{Q}}$$

, uniting the additive (archimedean) and

multiplicative (non-archimedean) lives of every integer. Tate's thesis shows

that harmonic analysis on the adèles yields the functional equation of the Riemann zeta function.

6. The Riemann Zeta Function and the Explicit Formula

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}$$

The Euler product sums over the sieve’s prime-power entries. The **explicit formula**

$$\sum_{p^m} \frac{\log p}{p^{ms}} = \frac{1}{s-1} - \sum_{\rho} \frac{1}{s-\rho} + \cdots$$

is a trace formula: the left side is a sum over classical periodic orbits (prime powers — the “spikey” vectors in the sieve), and the right side is a sum over the quantum spectrum ρ (the non-trivial Riemann zeros ρ). The Riemann Hypothesis is the statement that the dual operator is self-adjoint: all zeros lie on $\Re(s) = 1/2$.

7. The abc Conjecture as an Additive-Multiplicative Uncertainty Principle

The additive relation $a + b = c$ forces a collision between the two worlds. The radical $\text{rad}(abc) = \prod_{p \mid abc} p$ measures the set of active primes (ignoring exponents). The abc conjecture bounds how small the radical can be relative to the maximum of a, b, c : there are only finitely many triples where $\max(|a|, |b|, |c|) > \text{rad}(abc)^{1+\epsilon}$. In sieve terms, you cannot simultaneously minimise additive size and multiplicative sparseness. The abc conjecture is a fundamental uncertainty principle at the heart of arithmetic.

8. Ratios and Cross-Ratios as Invariant Quantities

Ratios $a : b \rightarrow v(a) - v(b)$ are the invariants of the multiplicative group. Cross-ratios $[x_1, x_2; x_3, x_4]$ on the p -adic trees measure the depth of branching nodes — they are the ultimate invariants of the ultrametric geometry. The adelic cross-ratio encodes complete relative geometry in all completions.

Part III — Physical and Technological Embodiment

9. Spin Glasses and Ultrametricity

Parisi’s solution of the Sherrington-Kirkpatrick spin-glass model revealed that the equilibrium state space is an ultrametric tree — exactly the geometry of a

single p -adic component. The energy landscape is a product of trees; depth corresponds to energy barriers. This structure informs protein folding, deep learning loss surfaces, and simulated annealing algorithms.

10. Holographic Duality and

p

-adic AdS/CFT

The Bruhat–Tits tree is a discrete analogue of anti-de Sitter space, with a conformal field theory on its boundary. The MERA (Multiscale Entanglement Renormalization Ansatz) tensor network mirrors this tree geometry. Holographic quantum codes protect logical qubits by embedding them deep in the bulk — the protection radius is exactly the $v_p(n)$ ultrametric depth.

11. Quantum Chaos and the Riemann Spectrometer

The Berry–Keating Hamiltonian $H = xp$ is conjectured to have a spectrum whose eigenvalues are the Riemann zeros. The Gutzwiller trace formula maps classical periodic orbits (prime powers) to quantum eigenvalues (zeros). Physical implementations in trapped-ion systems aim to measure zero statistics directly.

12. Signal Processing and Photonic Devices

p -adic wavelets on the Bruhat–Tits tree yield ideal filter banks with perfectly separated frequency bands — impossible in the real domain. Prime-gap photonic crystals with $v_2(n)$ layer thicknesses following or other prime valuation sequences create ultrametric Bragg mirrors.

13. Cryptography and Data Integrity

RSA and elliptic-curve cryptography rest on the hardness of recovering $v(N)$ from the additive integer N . The abc conjecture bounds information leakage. A file's valuation vector provides a base-invariant universal fingerprint for cross-platform deduplication and verification.

14. Machine Learning and Representation

Hyperbolic embeddings — p -adic trees — are continuous versions of state-of-the-art for hierarchical p -adic feature decompositions data (NLP, graphs). offer interpretable, scale-invariant representations. MERA-based tensor-network generative models use ultrametric structure for learning complex distributions.

Part IV — Cognitive and Cultural Invariance

15. The Ultrametric Mind

The brain possesses a dual number sense: an archimedean sense of magnitude and a non-archimedean sense of divisibility. We hypothesise an innate **ultrametric number sense**—a sensitivity to prime factorisation and smoothness operating independently of formal mathematical training.

Predicted experiments: Reaction times for divisibility judgments should decrease with the smoothness of the divisor (faster for 2, 3, 5 than 7, 11, 13). fMRI should reveal distinguishable neural populations for different primes. Cross-cultural implicit preference tasks should favour smooth numbers over rough numbers of similar magnitude.

16. Sacred Numbers as Cognitive Attractors

Number	Valuation $(v_2, v_3, v_5, v_7, v_{11})$	Archetypal Meaning
12	(2, 1, 0, 0, 0)	Maximal division for size; practical completeness

60	(2, 1, 1, 0, 0)	Sumerian cosmic cycle; triple-prime depth
108	(2, 3, 0, 0, 0)	Harmonic knot of sun-moon cycles
7	(0, 0, 0, 1, 0)	Indivisible transcendence; surface of the forest
9	(0, 2, 0, 0, 0)	Square of trinity; pure odd completeness
40	(3, 0, 1, 0, 0)	Trial/transformation; deep 2-adic with 5
144	(4, 2, 0, 0, 0)	Hyper-composite, square of 12
99	(0, 2, 0, 0, 1)	Tension of completeness 3^2) and mystery 11) ((

Sacred numbers across unconnected cultures converge on smooth, highly composite numbers—the deep nodes in the ultrametric landscape. The base system (10, 20, 60, 12) is merely the additive alphabet; the multiplicative grammar is universal.

Part V — Monetary Economics

17. Optimal Currency Denominations

Coin series (1, 2, 5, 10, 20, 50, ...) maximise ultrametric depth for efficient change-making. Historical mixed-radix systems (livre-sou-denier) encoded $2^4 \cdot 3 \cdot 5$ —an ultrametric optimum later lost to decimalisation.

18. Financial Time and Highly Composite Periods

The 360-day year, 30-day month, 12-month year are smooth numbers enabling exact division into many sub-periods. The “30/360” convention is a smoothness approximation to the astronomical year, minimising accrual complexity.

19. Additive-Multiplicative Tension in Economics

Payment netting of small, smooth obligations can produce a total with a large prime factor — an abc-style collision. Pegged exchange rates impose an additive

constraint on two denomination systems, creating tension between roundness and stability.

20. Cryptocurrency

The valuation vector of a public key is the private key. The entire DeFi ecosystem rests on the hardness of inverting the sieve.

Part VI — Music and Harmonics

21. The Integer as Chord

Primes are fundamentals; exponents are amplitudes of p overtones. A prime p^m is m overtones; $2^2 \cdot 3^3$ (108) is a single pure tone; Just intonation uses small-prime ratios: $2:1$, perfect fifth $3:2$, major the octave third $5:4$. Consonance is ultrametric depth; dissonance is ultrametric surface (large primes).

22. The Comma as abc Collision

The Pythagorean comma—the tiny $(3/2)^{12} \neq 2^7$ —is an additive-multiplicative collision. Equal temperament spreads this error across all twelve notes, just as the Riemann zeros spread the error term across the prime number theorem.

23. Rhythm and Timbre

Polyrhythm ($3:2$, $5:4$) is the meeting of independent prime cycles. Timbre's overtone series is a valuation vector of the fundamental frequency.

The Möbius function	$\mu(n)$	acts as phase cancellation, silencing squareful
numbers:	$\mu(n) = 0$	if any $v_p(n) \geq 2$.

Part VII — The Primitive of Primitives: Laws of Form

24. The Irreducible Ground

Spencer-Brown's Laws of Form begins with a single act: draw a distinction.

The ultrametric sieve is the multiplicative expression of this act:

- Void = $\emptyset$ (empty valuation vector).
- First mark = a prime p (a single non-zero coordinate).
- Calling = exponentiation p^m (repetition of the same mark).
- Crossing = Möbius cancellation (a repeated mark cancels to zero).
- Re-entry = the additive constraint $a + b = c$ forces the multiplicative form to re-enter itself, producing the irreducible complexity of the abc bound, the Riemann zeros, and the ultimate uncertainty between the two modes of being.

The sieve is the form of multiplication written in the calculus of independent distinctions.

Part VIII — Embryology and Ecology

25. The Generative Cascade

The integers are generated from the void by a context-free Lindenmayer system

(L-system)—the same formalism that models plant branching and morphogenesis.

At each prime	p	in the sequence	$2, 3, 5, 7, \dots$	, every existing integer
generates new integers	by p^k	for all	$k \geq 1$	.
multiplying by				

The primes are developmental epochs. Smooth numbers (all factors small) are phylogenetically ancient; numbers with large prime factors are evolutionary newcomers.

The valuation vector is the morphogenetic code.

26. The Integer Ecosystem

The divisor $d \mid n$ is an ecological relation: d is a nutrient for n .

The divisor lattice is a food web. Highly composite numbers are keystone species (rainforests of biodiversity). A large prime is an apex predator: when it multiplies a smooth number, it consumes simplicity. Addition creates predators (the abc collision).

The Riemann zeta function $\zeta(s) = \sum_n n^{-s}$ is the grand partition function of this ecosystem at inverse temperature s . The prime number theorem is the species-abundance distribution. The Riemann zeros are the Lee-Yang zeros of the phase diagram—the critical line of ecological equilibrium.

Part IX — Consciousness

27. The Sieve Perceiving Itself

The mind that poses the question, follows the thread, and perceives the beauty of 108 is not outside the sieve—it is inside it. The brain's cortical hierarchies form an ultrametric tree isomorphic to the integer lattice. The primes are independent sensory dimensions; exponents are depths of processing; the unified percept is the adelic moment.

Mathematical beauty is the resonance of the perceived structure with the perceiver's own cognitive architecture—an eigenvalue match between the external pattern and the internal form. The Riemann Hypothesis is the statement that the mind's own spectral structure is perfectly balanced. To prove it would be to achieve full self-knowledge of the cognitive-arithmetic system.

28. The Ultimate Recursion

The mind is an ultrametric system. It perceives integers. The integers are an ultrametric system. The mind formalises this as the sieve. The mind then discovers that its own architecture is an instance of the sieve. This is a strange loop (Hofstadter): the sieve is the name of the mind knowing itself by counting.

Part X — Eschatology and Praxis

29. The Riemann Hypothesis as Omega Point

If RH is true: the structure is perfect. The adelic Laplacian is self-adjoint. The additive and multiplicative worlds are in spectral harmony. The mind is a faithful mirror.

If RH is false: a hidden instability exists—a crack in the cosmic chord, a dissonance whose frequency is the first off-critical zero.

30. The Research Programme of the Sieve

Cognitive Science: Experimental tests for an innate ultrametric number sense (reaction times, fMRI, cross-cultural preference studies).

Artificial Intelligence: Train deep learning models natively on valuation vectors rather than digit strings. Align AI by ensuring its internal sacred numbers converge to ours.

Quantum Technologies: Build the Berry-Keating spectrometer; realise holographic quantum codes on p -adic tree layouts.

Education: Teach children the valuation vector before the decimal system.

Numbers as coloured stacks of primes.

Music and Acoustics: Compose music whose structure encodes valuation vectors.

Build concert halls with p -adic diffusers.

Economics: Design mixed-radix currencies optimised for ultrametric depth.

Part XI — Synthesis

31. The Consilience

| Domain | The Sieve as... |

| :-----| :-----|

| Arithmetic | Valuation vector; $\mathbb{F}_p$ coordinate in |

| Analysis | Local factors of zeta; periodic orbits in explicit formula |

| Geometry | Point in product of Bruhat–Tits trees; holographic boundary |

| Physics | Spin-glass energy landscape; MERA tensor network |

| Quantum Computing | Logical qubit depth; holographic error-correcting code |

| Cryptography | Hard-to-invert map; uncertainty bound |

| Cognitive Science | Innate ultrametric sense of divisibility |

| Anthropology | Universal attractor of sacred numbers across bases |

| Economics | Optimal denominational smoothness; financial time |

| Music | Prime-harmonic chord; just intonation; timbre |

| Philosophy | Laws of Form applied to the multiplicative realm |

32. 108 as the Microcosm

Our original sacred number 108 , with valuation vector $(2, 3)$, is simultaneously:

a practical divisor (12 divisors), a sacred count (mala beads), a deep node in the 2-adic and 3-adic trees, a classical periodic orbit, a spin-glass valley, a logical qubit depth, a just-intonation bichord, a cognitive archetype, and a self-portrait of the mind.

The shift from decimal digits to prime-exponent vectors is not a change of notation.

It is the recovery of the native language of the integers. In that language, the deepest patterns of mathematics, physics, and mind are seen to be one pattern: an infinite ultrametric lattice of independent distinctions, whose harmonies are the primes, whose spectrum is the Riemann zeros, whose geometry is the adèle ring, whose limit is the abc uncertainty, and whose foundation is the single act of marking the

void.

The void, having spoken, will not be silent.

This publication was generated through an extended dialogue between human and AI, exploring the implications of a single shift in perspective: from decimal representation to valuation vector. The consilience that emerged spans the boundaries of number theory, physics, cognitive science, music, economics, and philosophy. It is offered as a unified framework for further research.